INTERNSHIP FINAL PROJECT REPORT • WEB DEVELOPMENT

# CLAIM PRO — Claim Processing System

Comprehensive Single-Page Architecture & Lifecycle Processing Report

**Organization:** Data Alcott Systems | **Task ID:** WD-INS-002
**Student Code:** DAS-INS-002 | **Domain:** Web Development Internship
**Submission:** October 2026 | **Reference Portal:** freeinternships.in/blog

**ARCHITECTURE SCOPE & DEMONSTRATION NOTICE:** This project is an authentic **pure frontend-only demonstration** designed strictly using HTML5, CSS3, and vanilla JavaScript (ES6+). State persistence is maintained exclusively through browser **localStorage** and active session Object URLs. It **does not include a production backend, server environment, or relational database**.

## 1 PROJECT OBJECTIVE

The primary objective of this internship project is to architect, develop, and thoroughly validate **"CLAIM PRO — Claim Processing System"**, an enterprise-grade, responsive Single-Page Application (SPA). The application streamlines the complete end-to-end lifecycle of insurance claims—covering digital intake filing, multi-tier document validation, officer review queues, formal approval/rejection decisioning, financial settlement disbursements, and actuarial reporting. Operating strictly without heavy external web frameworks (such as React, Vue, or Angular) or server-side runtimes, the system guarantees instant sub-second reactivity, dynamic client-side filtering, and persistent state across page reloads.

## 2 TECHNOLOGIES USED & ARCHITECTURE

### Semantic HTML5

Standards-compliant semantic markup utilizing `<header>`, `<nav>`, `<aside>`, `<main>`, `<section>`, `<article>`, and native `<dialog>` elements with accessible ARIA roles.

### Vanilla CSS3 Design System

Tokenized custom properties (`--primary: #7C3AED`, `--bg-app: #F8FAFC`), responsive CSS Grid and Flexbox layouts (320px–1440px), high-contrast Dark Mode (`data-theme="dark"`), and `@media print` rules.

### Vanilla JavaScript (ES6+)

Modular Object-Oriented Store architecture, asynchronous DOM updates, active session Blob URL generation, secure CSV generation with formula injection defense, and hash-based routing.

### localStorage & Pure SVG

Browser `localStorage` handles complete client-side data persistence across sessions. Pure mathematical inline SVG renders dynamic 6-month filing trends and calculated donut charts with 0 CDN bloat.

## 3 IMPLEMENTED FUNCTIONAL MODULES (10 CORE MODULES)

1. **App Shell & Navigation:** Collapsible desktop/mobile drawer, global search bar, notification center, and instant theme toggling.
2. **Executive Dashboard:** 10 dynamic KPI cards, inline SVG 6-month filing trend lines, SVG status donut chart, and live audit feed.
3. **Multi-Step Claim Wizard:** 4-step guided intake with live field validation, drag-and-drop document upload, and review confirmation.
4. **Master Claims Archive:** Searchable, sortable, multi-filter table with live status badges, category filters, and pagination.
5. **Claim Detail & Timeline Modal:** Deep audit modal containing complete claim info, document inspection, internal notes, and chronological logs.
6. **Review & Verification Queue:** Granular document verification auditing, rejection reason capture, and compliance guard enforcement.
7. **Settlement & Disbursement:** Financial payout processing, ceiling limit verification, payment channel dispatch, and printable voucher modal.
8. **Centralized Document Archive:** Unified document repository across all claims with instant search, status filtering, and preview links.
9. **Actuarial Reports & Data Export:** Status breakdown progress bars, category settlement matrices, and RFC-compliant CSV data exporters.
10. **RBAC & Profile Management:** Quick switching between Reviewer (Sarah Jenkins) and Claimant (Rahul Sharma) with data isolation and JSON backup/restore.

## 4 KEY BUSINESS LOGIC & ALGORITHMIC RULES

### COMPLIANCE Document Verification Approval Guard

To emulate rigorous regulatory standards, a claim **cannot transition to Approved** if any attached document is missing, pending, or rejected. The engine intercepts premature approval attempts and surfaces an inspection modal highlighting required audits.

### FINANCIAL Payout Ceiling & Settlement Sync

Disbursement amounts are strictly validated to be positive and cannot exceed the approved benefit limit. A claim transitions to terminal `Settled` state only once the payment status is marked `Paid` (remains `Approved` during `Processing`).

### SECURITY Spreadsheet Formula Injection Defense

Client-side CSV exports implement RFC-compliant quoting and proactively neutralize formula injection vectors by prepending a single quote (`'`) to any text value starting with `=`, `+`, `-`, `@`, `\t`, or `\r`.

### INTEGRITY Reactive Metric & State Integrity

All dashboard KPI totals, approval ratios, settlement sums, and chart geometry are recalculated in real time directly from underlying records on every mutation. Zero static counters or hardcoded values are utilized.

## 5 VERIFICATION, QUALITY ASSURANCE & TEST RESULTS

**Syntax & Static Analysis** PASS (0 Errors)

Validated with Node.js native engine:

- `node --check script.js`: 0 syntax errors, valid ES6+ parsing.
- Zero undeclared global mutations; strict block scoping verified.
- All event listeners cleanly registered on DOMContentLoaded.

**Automated Node.js Suites** 52 / 52 PASSED

- **DOM & Theme Integrity:** Validated required IDs, colors (#7C3AED, #EDE7FF), dark mode tokens.
- **Security Vectors:** CSV sanitizer verified against `=SUM()`, `+cmd`, `@exec`.
- **Workflow Gates:** Verified doc-approval block and settlement disbursement ceilings.
- **RBAC Isolation:** Verified claimant privacy vs reviewer master queue access.

**Headless Chrome Verification** DOM VERIFIED

Tested via Chrome CLI (`--headless --dump-dom`):

- Page load triggers state initialization: 8 seeded claims, ₹19,18,500 total volume.
- SVG trend lines & donut arcs render with dynamic geometry.
- Hash routes (`#claim-review`, `#settlements`, `#reports`) fully functional.

*Note on Testing Scope:* While automated unit and DOM assertions achieved 100% pass rates, interactive desktop browser walkthroughs (manual mouse clicks, modal transitions, and visual reviews) remain student-verified in live desktop browsers.

## 6 KEY ENGINEERING CHALLENGES & TECHNICAL SOLUTIONS

| Challenge Encountered | Observed Technical Constraint | Architectural Solution Implemented |
|---|---|---|
| **localStorage Quota Limits** | Converting user-uploaded PDF/image attachments directly to Base64 exceeds browser storage quotas (~5MB) across multiple claims. | Stored document metadata (name, size, type, timestamp, status) in `localStorage` while generating temporary active session `URL.createObjectURL` references for live visual previews. |
| **Premature Claim Approval** | Without server validation, reviewers could inadvertently approve claims prior to reviewing critical invoices and diagnostic bills. | Engineered an interception guard in the approval flow scanning supporting documents. If any item is pending/rejected, approval halts and presents a detailed resolution modal. |
| **Static Role Isolation (RBAC)** | Simulating authentic multi-tenant roles without an active authentication server or database backend. | Constructed a reactive user context controller that dynamically filters the claims archive by claimant name/email when in Claimant mode, while disabling officer-only queues. |

## 7 Key Learning Outcomes

- **Modern Vanilla JS Architecture:** Deepened mastery of modular state design, centralized data stores, and event-driven rendering without framework dependencies.
- **Accessible UI Patterns:** Implemented keyboard focus trapping, native ESC modal dismissal, and ARIA live regions for screen reader compliance.
- **Data Defense & Sanitization:** Learned formula injection defense in browser-generated spreadsheets and defensive JSON schema parsing.
- **Pure SVG Data Visualization:** Built dynamic coordinate mapping algorithms to render mathematical trend lines and donut charts.

## 8 Limitations & Production Roadmap

- **Production Backend Integration:** Client-side `localStorage` should be transitioned to a secure REST/GraphQL API backed by PostgreSQL and encrypted S3/GCS buckets for immutable compliance archiving.
- **Payment Gateway Integration:** The current settlement engine provides simulated disbursements; production deployment would incorporate Razorpay/Stripe webhooks or direct NEFT/ACH bank APIs.
- **Automated OCR Verification:** Integrating OCR models (e.g. Tesseract/Google Vision) to automatically extract hospital invoice totals and match them against filed amounts.

## 9 CONCLUSION & PROJECT EVALUATION

The **CLAIM PRO — Claim Processing System** has been fully designed, implemented, verified, and documented in strict adherence to the requirements outlined for internship task **WD-INS-002** under **Data Alcott Systems**. The resulting Single-Page Application provides a rich, responsive, and robust insurance lifecycle simulation demonstrating state persistence, business validation guards, and data export security. All source deliverables —including HTML5 markup, CSS3 styling, vanilla JavaScript logic, test suites, and documentation—are organized, version-controlled, and ready for local demonstration and static deployment.

**Project Status: Completed & Fully Verified** **Quality: 52/52 Tests Passing • Zero Syntax Errors** **Deployment: Static Web Ready**

## Figure A1: Claims Executive Dashboard & Analytics Overview

High-level KPI metric cards, claims volume statistics, 6-month filing trends, and status distribution donut visualization.

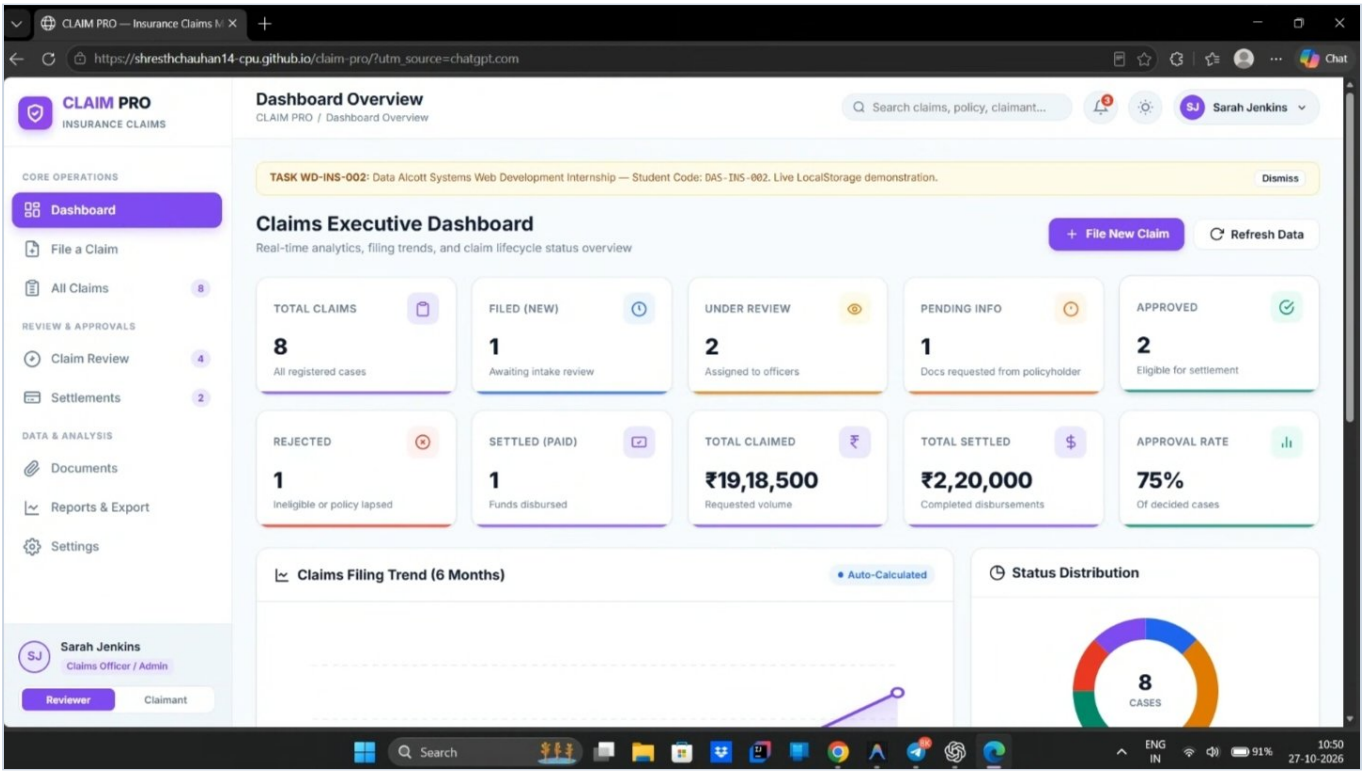



Figure A1 Caption: Source File: IMG_20261027_131534.jpg | Desktop Interface (1444 × 812 px)

## Figure A2: Mobile Responsive Dashboard — Core KPI Metric Cards

Adaptive mobile layout displaying Total Claims, Filed, Under Review, Pending Info, Approved, and Rejected counters.

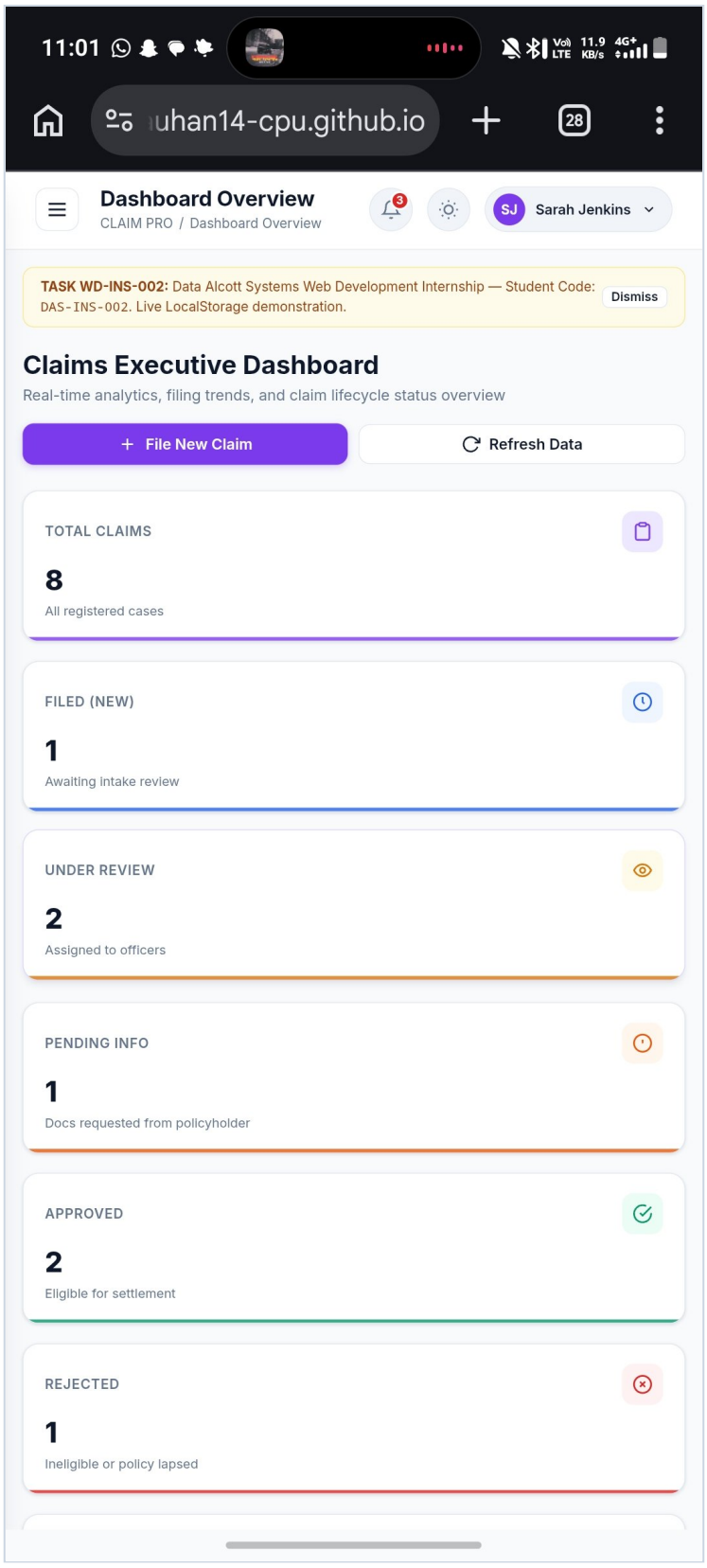



Figure A2 Caption: Source File: Screenshot_20261027_110104.jpg | Mobile Viewport (1080 × 2408 px)

## Figure A3: Mobile Responsive Dashboard — Financial Volume & Trend Charts

Settled amounts, total claimed volume (■19,18,500), approval rate gauge, and dynamic SVG trend line on mobile viewport.

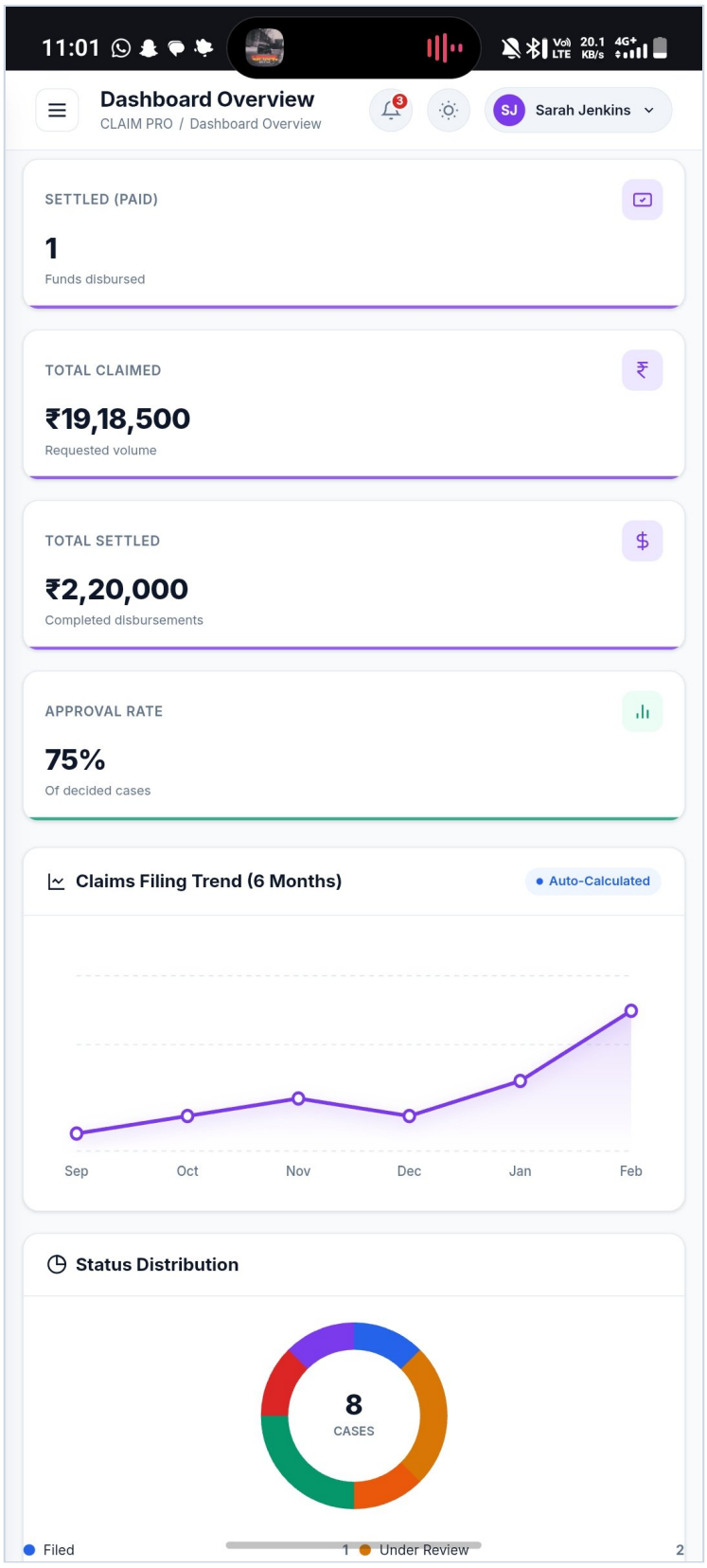



Figure A3 Caption: Source File: Screenshot_20261027_110109.jpg | Mobile Viewport (1080 × 2408 px)

## Figure A4: Mobile Responsive Dashboard — Recent Claims Queue & Live Activity

Compact mobile table displaying recent claim records, policy numbers, claim amounts, and live audit event log.

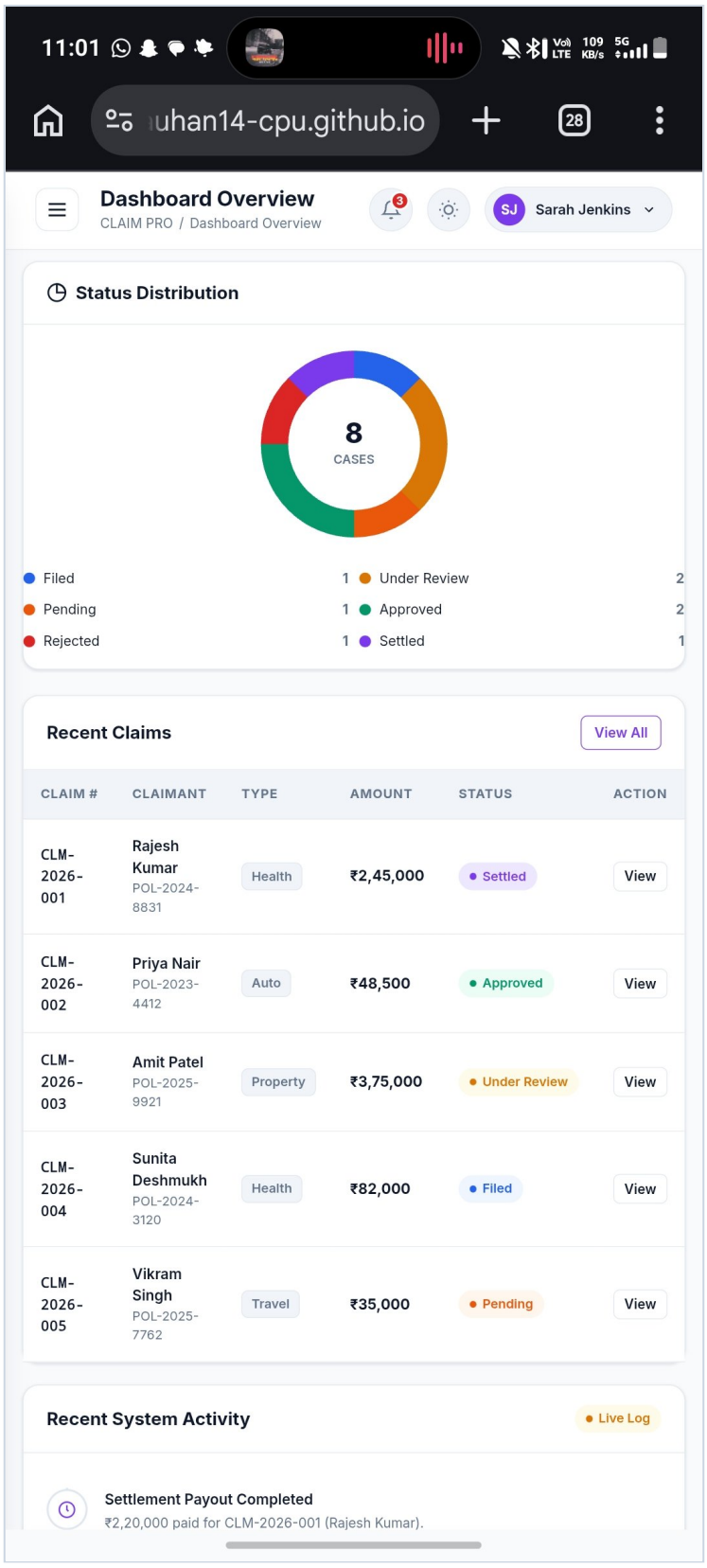



Figure A4 Caption: Source File: Screenshot_20261027_110114.jpg | Mobile Viewport (1080 × 2408 px)

## Figure A5: Mobile Interface — Off-Canvas Navigation Drawer & Role Switcher

Slide-out navigation menu showing operational views, notification badges, active reviewer profile, and one-tap role toggle.

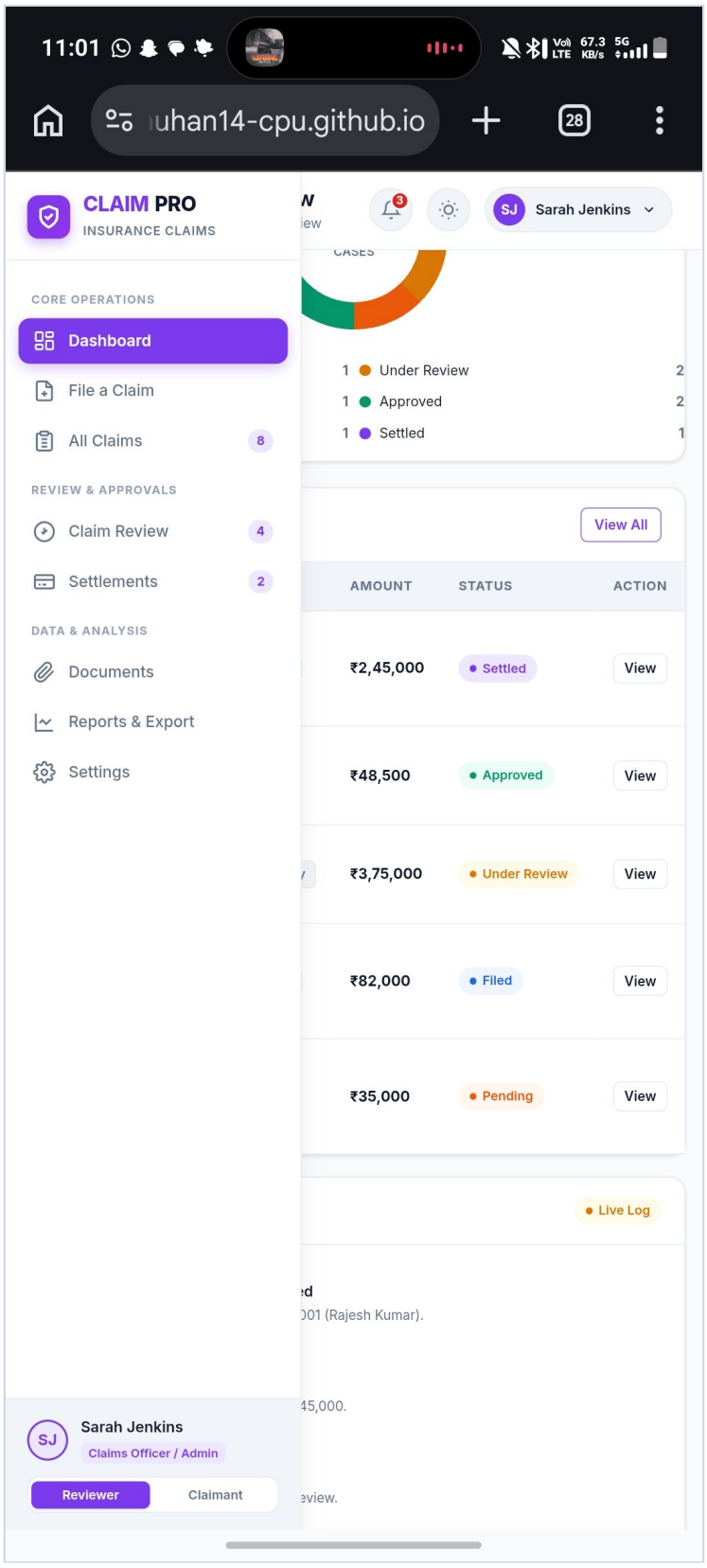



Figure A5 Caption: Source File: Screenshot_20261027_110118.jpg | Mobile Viewport (1080 × 2408 px)

## Figure A6: Mobile Interface — Responsive Navigation Menu (Expanded State)

Off-canvas navigation view highlighting core operation links, document repository access, and system settings navigation.

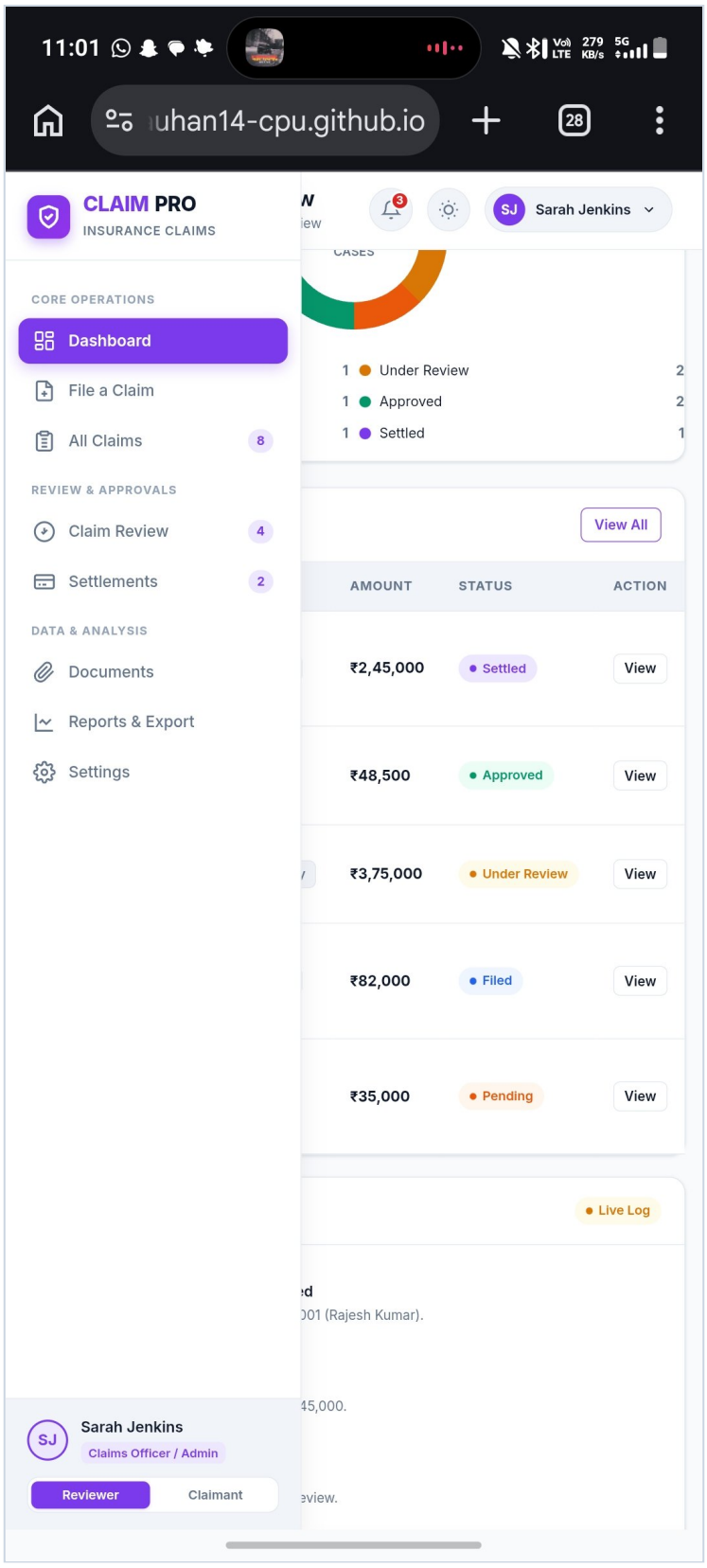



Figure A6 Caption: Source File: Screenshot_20261027_110122.jpg | Mobile Viewport (1080 × 2408 px)

## Figure A7: Mobile Claim Filing Portal — Step 1: Policyholder & Claimant Details

Step 1 of the 4-step intake workflow with responsive form inputs, policy identification, and validation controls.

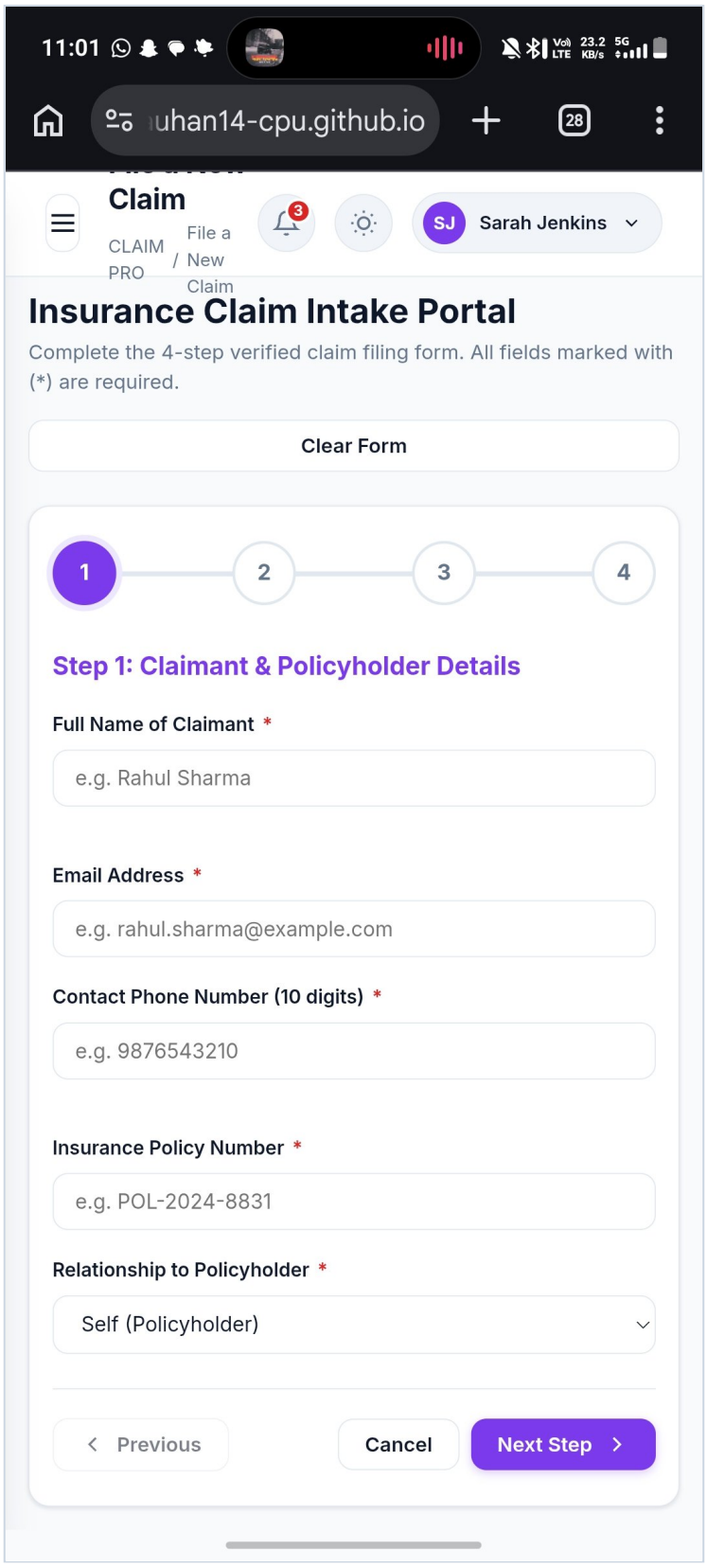



Figure A7 Caption: Source File: Screenshot_20261027_110132.jpg | Mobile Viewport (1080 × 2408 px)

## Figure A8: Mobile Claims Directory — Search, Filter & Master Archive

Complete mobile claims registry featuring multi-parameter search, status filtering, category sorting, and paginated records.

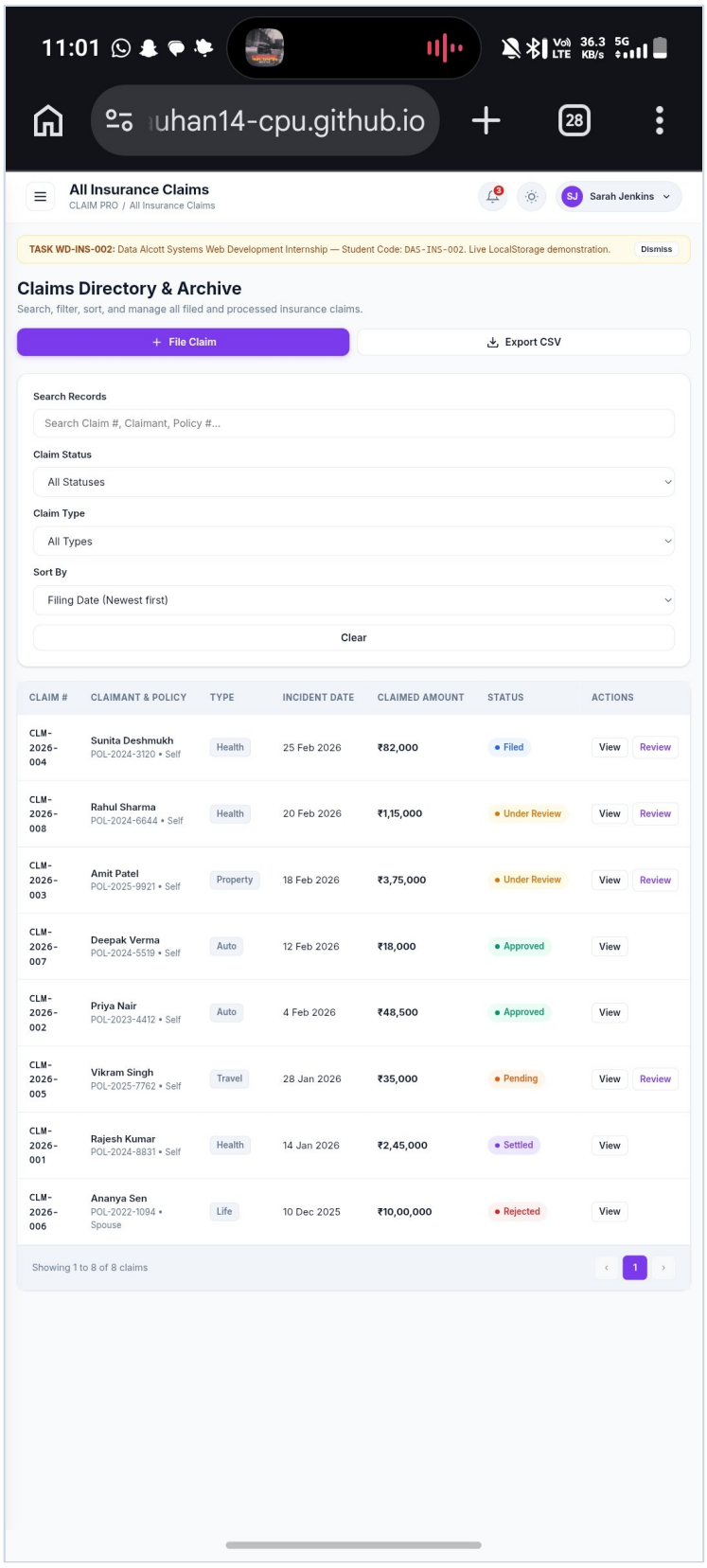



Figure A8 Caption: Source File: Screenshot_20261027_110141.jpg | Mobile Viewport (1080 × 2408 px)

## Figure A9: Desktop System Settings & LocalStorage Management

RBAC simulation profile configuration, reviewer/claimant permission controls, UI theme preferences, and data persistence controls.

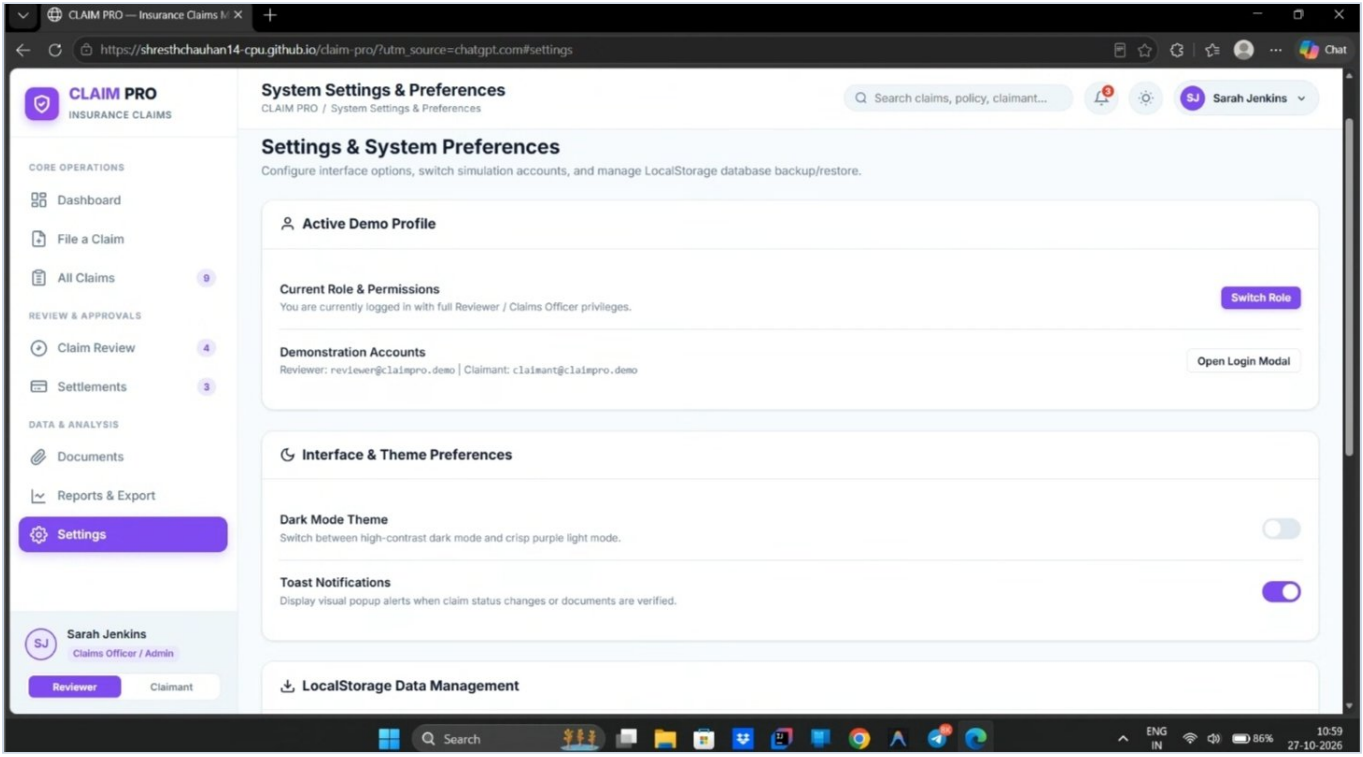



Figure A9 Caption: Source File: IMG_20261027_130919.jpg | Desktop Interface (1452 × 800 px)

## Figure A10: Desktop Analytical Reports & Actuarial Lifecycle Breakdown

Executive management report presenting status phase distribution, claimed liabilities, distribution progress gauges, and CSV export.

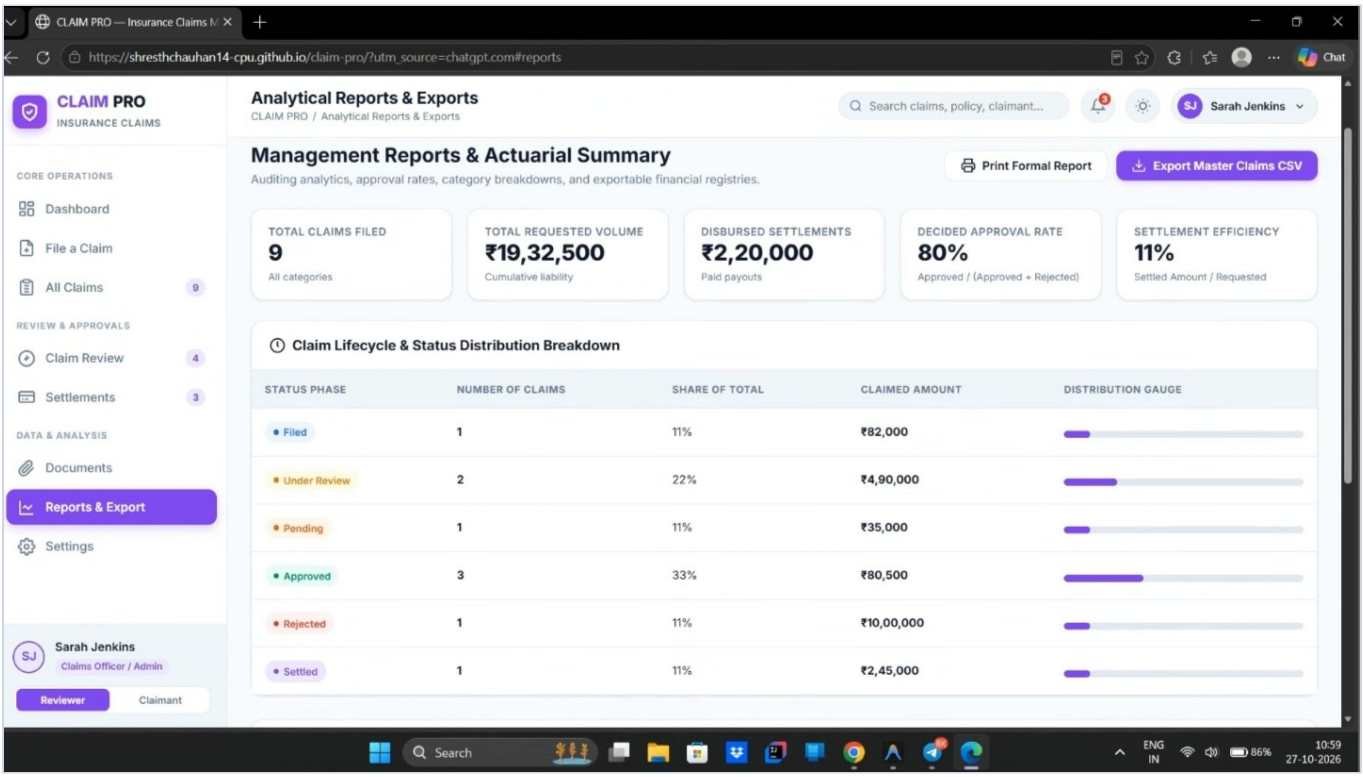



Figure A10 Caption: Source File: IMG_20261027_130956.jpg | Desktop Interface (1440 × 812 px)

## Figure A11: Central Document Library & Verification Management

Comprehensive document repository displaying file formats, upload sizes, verification status badges, and preview modals.

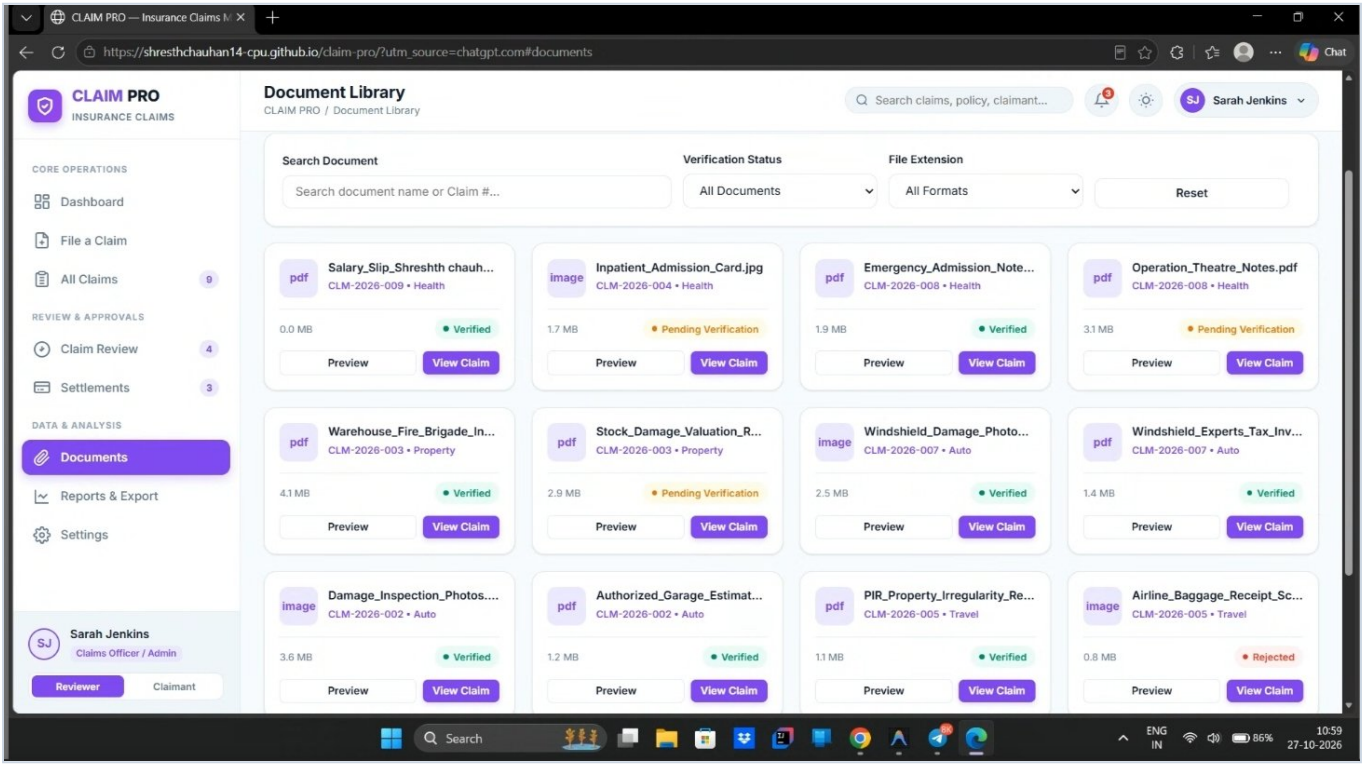



Figure A11 Caption: Source File: IMG_20261027_131028.jpg | Desktop Interface (1456 × 812 px)

## Figure A12: Settlement Processing & Disbursement Management

Payout queue for approved claims, pending liability balances, NEFT/direct bank transfer execution, and transaction history.

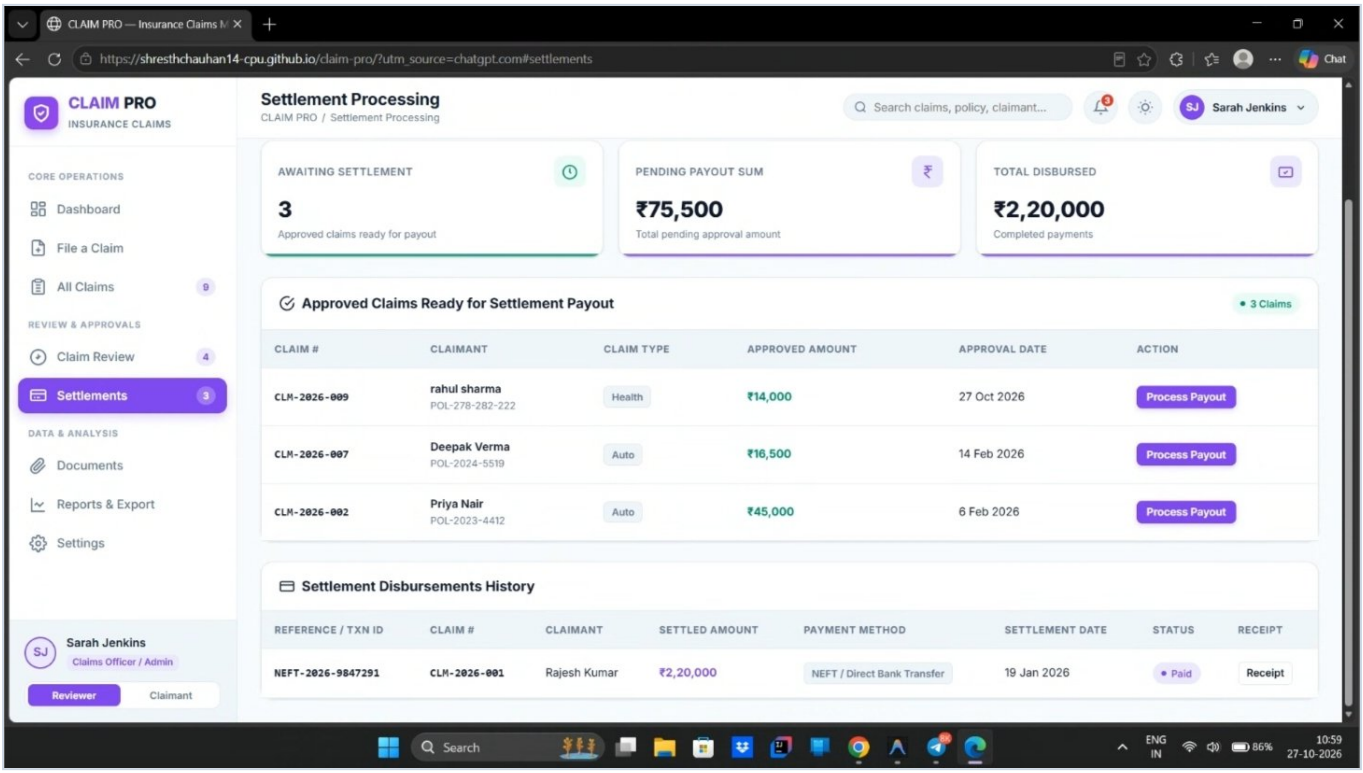



Figure A12 Caption: Source File: IMG_20261027_131103.jpg | Desktop Interface (1452 × 816 px)

## Figure A13: Officer Review & Document Verification Queue

Claims adjudication workspace featuring per-document validation controls, verification checklists, and audit action gates.

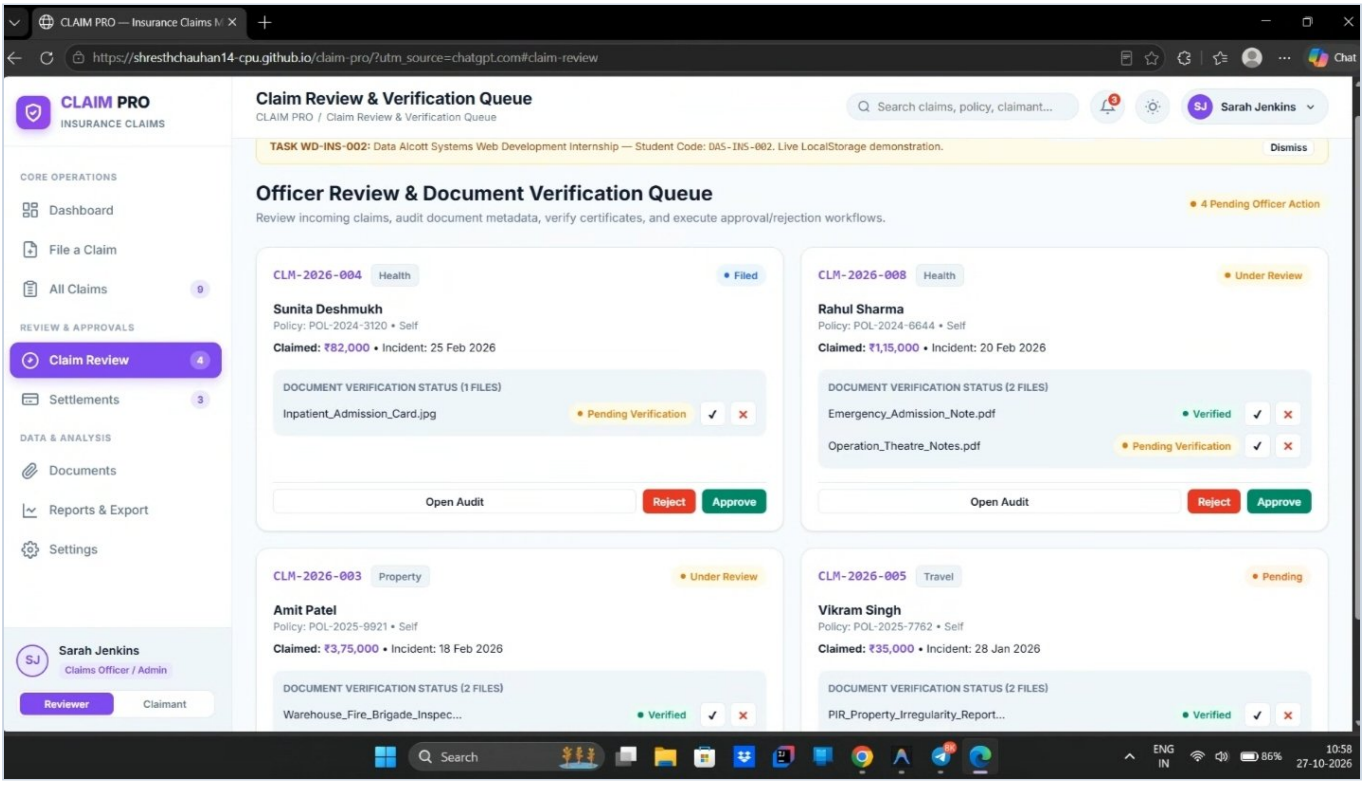



Figure A13 Caption: Source File: IMG_20261027_131140.jpg | Desktop Interface (1432 × 816 px)

## Figure A14: Claims Directory & Archive with Interactive Toast Notification

Master claims archive table showcasing real-time feedback notifications ("Claim Approved — CLM-2026-009") upon action execution.

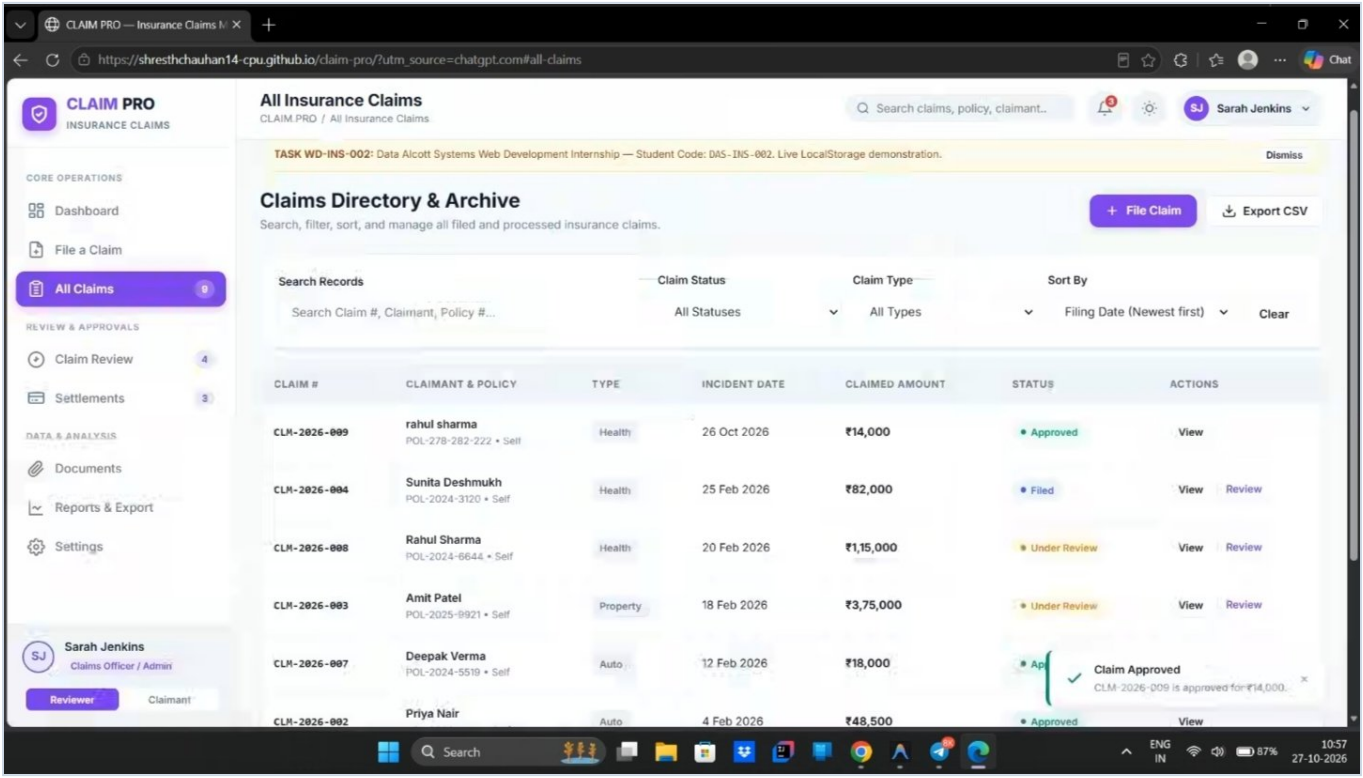



Figure A14 Caption: Source File: IMG_20261027_131217.jpg | Desktop Interface (1444 × 820 px)

## Figure A15: Claim Settlement Approval Modal & Verification Gate

Adjudication modal enforcing document verification prerequisites, approval amount capping, and mandatory officer remarks.

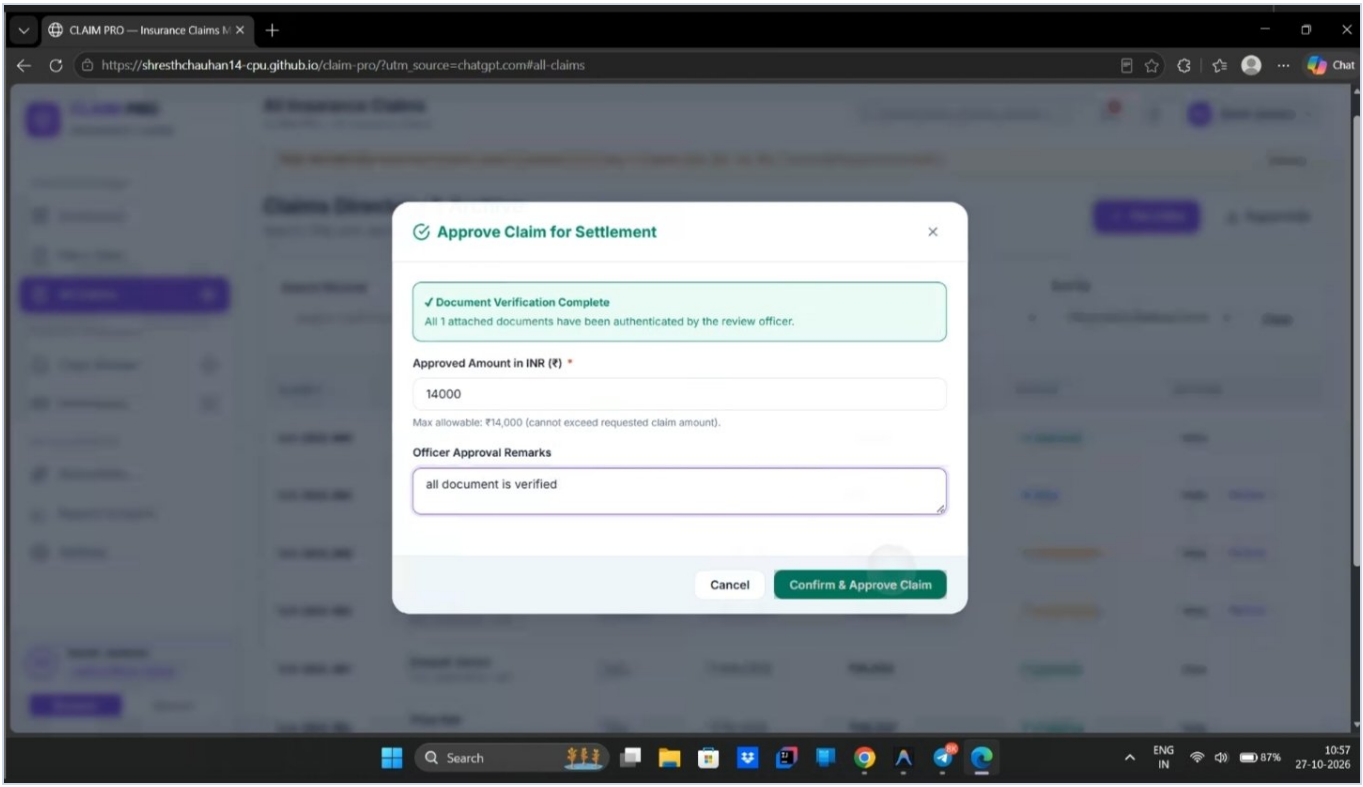



Figure A15 Caption: Source File: IMG_20261027_131302.jpg | Desktop Interface (1444 × 828 px)

## Figure A16: Claim Details & Supporting Document Audit Dialog

Interactive modal showing claim summary, incident statement, attached evidence checklist, and real-time verification status.

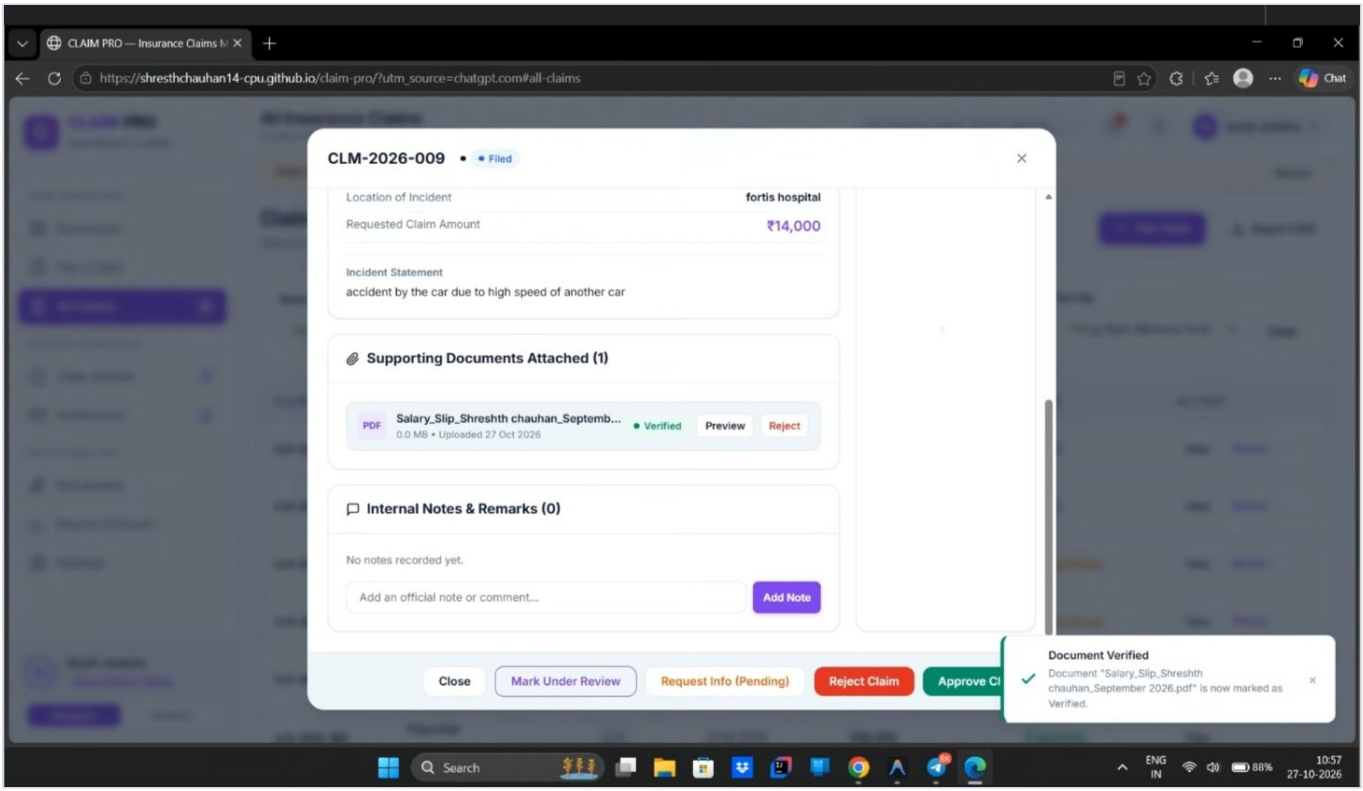



Figure A16 Caption: Source File: IMG_20261027_131404.jpg | Desktop Interface (1452 × 836 px)

## Figure A17: Multi-Step Claim Intake — Step 4: Final Review & Submission

Consolidated intake summary displaying policyholder data, financial amounts, incident narrative, and legal affirmation checkbox.

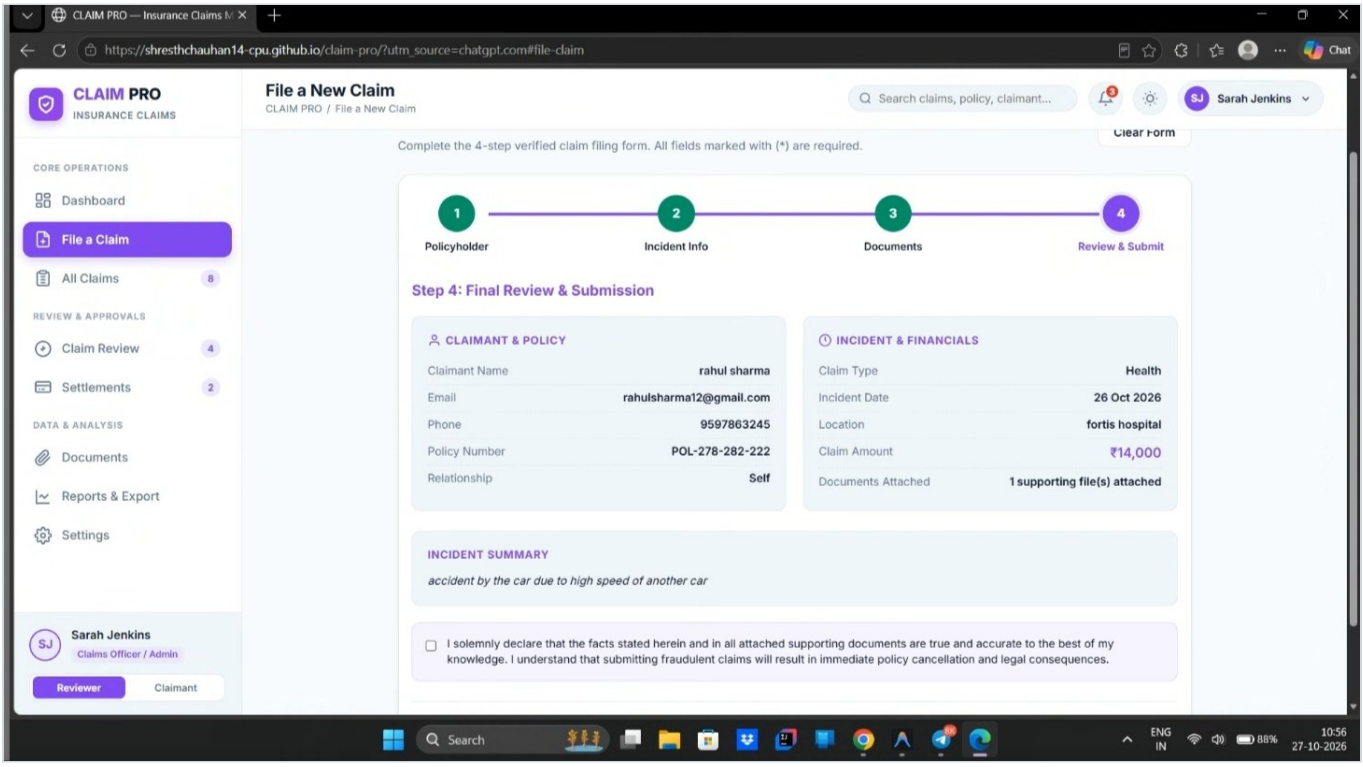



Figure A17 Caption: Source File: IMG_20261027_131446.jpg | Desktop Interface (1452 × 808 px)